№2

:

2026-03-19

|

1

1.

1.1

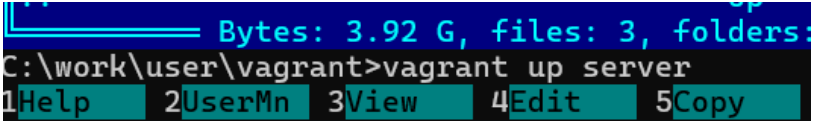
-
-
- Markdown

2

2.

2.1 DNS-

- bahis
- sudo -i
- root



1: sudo -i

- bind bind-utils
- dnf -y install
-

```
[user@server.user.net ~]$ sudo -i
[sudo] password for user:
[root@server.user.net ~]# dnf -y install bind bind-utils
Last metadata expiration check: 0:23:11 ago on Tue 06 Jan 2026 01:18:33 PM UTC.
Package bind-utils-32:9.16.23-18.el9_4.1.x86_64 is already installed.
Dependencies resolved.
================================================================================================================
 Package                        Architecture       Version                          Repository            Size
================================================================================================================
Installing:
 bind                           x86_64             32:9.16.23-34.el9_7.1            appstream            488 k
Upgrading:
 bind-libs                      x86_64             32:9.16.23-34.el9_7.1            appstream            1.2 M
 bind-license                   noarch             32:9.16.23-34.el9_7.1            appstream             13 k
 bind-utils                     x86_64             32:9.16.23-34.el9_7.1            appstream            199 k
 openssl                        x86_64             1:3.5.1-4.el9_7                  baseos               1.4 M
 openssl-devel                  x86_64             1:3.5.1-4.el9_7                  appstream            3.4 M
 openssl-libs                   x86_64             1:3.5.1-4.el9_7                  baseos               2.3 M
Installing dependencies:
 bind-dnssec-doc                noarch             32:9.16.23-34.el9_7.1            appstream             45 k
 openssl-fips-provider          x86_64             1:3.5.1-4.el9_7                  baseos               812 k
 python3-bind                   noarch             32:9.16.23-34.el9_7.1            appstream             61 k
 python3-ply                    noarch             3.11-14.el9.0.1                  baseos               103 k
Installing weak dependencies:
 bind-dnssec-utils              x86_64             32:9.16.23-34.el9_7.1            appstream            113 k

Transaction Summary
================================================================================================================
Install  6 Packages
Upgrade  6 Packages
```

2: bind bind-utils dnf

- dig www.yandex.ru
- NOERROR
- A-
-

```
Complete!
[root@server.user.net ~]# dig www.yandex.ru

; <<>> DiG 9.16.23-RH <<>> www.yandex.ru
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 58434
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;www.yandex.ru.                 IN      A

;; ANSWER SECTION:
www.yandex.ru.          3600    IN      A       5.255.255.77
www.yandex.ru.          3600    IN      A       77.88.44.55
www.yandex.ru.          3600    IN      A       77.88.55.88

;; Query time: 28 msec
;; SERVER: 10.0.2.3#53(10.0.2.3)
;; WHEN: Tue Jan 06 13:50:10 UTC 2026
;; MSG SIZE  rcvd: 79

[root@server.user.net ~]#
```

3: dig www.yandex.ru

2.2 DNS-

- /etc/resolv.conf
- bahis.net
- DNS-

```
[root@server.user.net ~]# sudo -i
[root@server.user.net ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search user.net
nameserver 10.0.2.3
[root@server.user.net ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search user.net
nameserver 10.0.2.3
[root@server.user.net ~]# cat /etc/named.conf
//
// named.conf
//
// Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
// server as a caching only nameserver (as a localhost DNS resolver only).
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//

options {
        listen-on port 53 { 127.0.0.1; };
        listen-on-v6 port 53 { ::1; };
        directory       "/var/named";
        dump-file       "/var/named/data/cache_dump.db";
        statistics-file "/var/named/data/named_stats.txt";
        memstatistics-file "/var/named/data/named_mem_stats.txt";
        secroots-file   "/var/named/data/named.secroots";
        recursing-file  "/var/named/data/named.recursing";
```

- /etc/named.conf
- recursion yes
- 127.0.0.1
- localhost

root@server:~

```
[root@server.user.net ~]# cat /var/named/named.ca

; <<>> DiG 9.18.20 <<>> -4 +tcp +norec +nostats @d.root-servers.net
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 47286
;; flags: qr aa; QUERY: 1, ANSWER: 13, AUTHORITY: 0, ADDITIONAL: 27

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1450
;; QUESTION SECTION:
;.                              IN      NS

;; ANSWER SECTION:
.                       518400  IN      NS      a.root-servers.net.
.                       518400  IN      NS      b.root-servers.net.
.                       518400  IN      NS      c.root-servers.net.
.                       518400  IN      NS      d.root-servers.net.
.                       518400  IN      NS      e.root-servers.net.
.                       518400  IN      NS      f.root-servers.net.
.                       518400  IN      NS      g.root-servers.net.
.                       518400  IN      NS      h.root-servers.net.
.                       518400  IN      NS      i.root-servers.net.
.                       518400  IN      NS      j.root-servers.net.
.                       518400  IN      NS      k.root-servers.net.
.                       518400  IN      NS      l.root-servers.net.
.                       518400  IN      NS      m.root-servers.net.
```

5: /etc/named.conf

- named.ca
- NS, A, AAAA-

```
[root@server.user.net ~]# cat /var/named/named.localhost
$TTL 1D
@       IN SOA  @ rname.invalid. (
                                        0       ; serial
                                        1D      ; refresh
                                        1H      ; retry
                                        1W      ; expire
                                        3H )    ; minimum
        NS      @
        A       127.0.0.1
        AAAA    ::1
[root@server.user.net ~]#
```

6: named.ca ( )

- named.localhost
- localhost

```
[root@server.user.net ~]# systemctl start named
[root@server.user.net ~]# systemctl enable named
Created symlink /etc/systemd/system/multi-user.target.wants/named.service → /usr/lib/systemd/system/named.service.
[root@server.user.net ~]#
```

7: named.localhost

- named.loopback
- 127.0.0.1

root@server:~

```
[root@server.user.net ~]# dig www.yandex.ru

; <<>> DiG 9.16.23-RH <<>> www.yandex.ru
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 62703
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;www.yandex.ru.                 IN      A

;; ANSWER SECTION:
www.yandex.ru.          3600    IN      A       5.255.255.77
www.yandex.ru.          3600    IN      A       77.88.55.88
www.yandex.ru.          3600    IN      A       77.88.44.55

;; Query time: 4 msec
;; SERVER: 10.0.2.3#53(10.0.2.3)
;; WHEN: Tue Jan 06 14:05:38 UTC 2026
;; MSG SIZE  rcvd: 79

[root@server.user.net ~]#
```

8: named.loopback

- `systemctl start named  enable named`
- `dig`
-

```
[user@server.user.net ~]$ nmcli connection edit eth0

===| nmcli interactive connection editor |===

Editing existing '802-3-ethernet' connection: 'eth0'

Type 'help' or '?' for available commands.
Type 'print' to show all the connection properties.
Type 'describe [<setting>.<prop>]' for detailed property description.

You may edit the following settings: connection, 802-3-ethernet (ethernet), 802-1x, dcb, sriov, ethtool, match, ipv4,
 ipv6, hostname, link, tc, proxy
nmcli> romove ipv4.dns
Unknown command: 'romove ipv4.dns'
nmcli> remove ipv4.dns
nmcli> set ipv4.ignore-auto-dns yes
nmcli> ipv4.dns 127.0.0.1
Unknown command: 'ipv4.dns 127.0.0.1'
nmcli> save
Connection 'eth0' (52b757e4-3833-4753-b0eb-de0af7f26f57) successfully updated.
nmcli> quit
[user@server.user.net ~]$
```

9: named dig

- `nmcli` DNS 127.0.0.1
- auto-DNS
- `/etc/resolv.conf`

```
[root@server.user.net ~]# systemctl restart NetworkManager
[root@server.user.net ~]#
[root@server.user.net ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search user.net
nameserver 127.0.0.1
[root@server.user.net ~]#
```

10: DNS nmcli resolv.conf

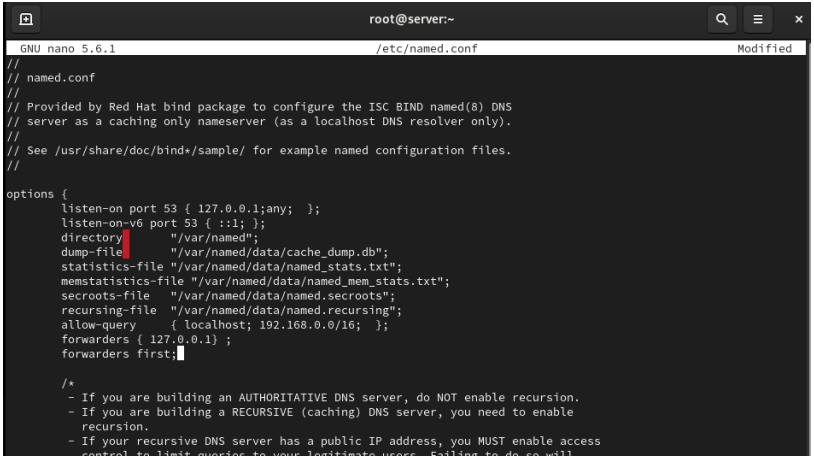
- named.conf listen-on {127.0.0.1; any;}
- 192.168.0.0/16
- firewall DNS

- 53
- `named` (UDP)

```
[root@server.user.net ~]# firewall-cmd --add-service=dns
success
[root@server.user.net ~]# firewall-cmd --add-service=dns --permanent
success
[root@server.user.net ~]# sof | grep UDP
bash: sof: command not found...
[root@server.user.net ~]# lsof | grep UDP
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1001/gvfs
      Output information may be incomplete.
lsof: WARNING: can't stat() fuse.portal file system /run/user/1001/doc
      Output information may be incomplete.
avahi-dae  666                        avahi   12u     IPv4              21552       0t0        UDP *:mdns
avahi-dae  666                        avahi   13u     IPv6              21553       0t0        UDP *:mdns
avahi-dae  666                        avahi   14u     IPv4              21554       0t0        UDP *:40540
avahi-dae  666                        avahi   15u     IPv6              21555       0t0        UDP *:54159
chronyd    700                       chrony    5u     IPv4              21318       0t0        UDP localhost:323
chronyd    700                       chrony    6u     IPv6              21319       0t0        UDP localhost:323
named     1043                        named    6u     IPv4              30297       0t0        UDP server.user.net:d
omain
named     1043                        named   65u     IPv4              22047       0t0        UDP localhost:domain
named     1043                        named   66u     IPv4              22048       0t0        UDP localhost:domain
named     1043                        named   67u     IPv4              22049       0t0        UDP localhost:domain
named     1043                        named   68u     IPv4              22050       0t0        UDP localhost:domain
named     1043                        named   69u     IPv4              22051       0t0        UDP localhost:domain
named     1043                        named   80u     IPv6              22057       0t0        UDP localhost:domain
```

12: 53 named

- forwarders {127.0.0.1;}
- forward first
- dnssec-enable dnssec-validation



13: forwarders DNSSEC named.conf

- named.rfc1912.zones
- bahis.net
- include named.conf

```
[user@server.user.net ~]$ cd /var/named
bash: cd: /var/named: Permission denied
[user@server.user.net ~]$
[user@server.user.net ~]$
[user@server.user.net ~]$ sudo -i
[sudo] password for user:
[root@server.user.net ~]# cd /var/named
[root@server.user.net named]# mkdir -p /var/named/master/fz
[root@server.user.net named]# mkdir -p /var/named/master/rz
[root@server.user.net named]# ls
data  dynamic  master  named.ca  named.empty  named.localhost  named.loopback  slaves
[root@server.user.net named]#
```

14: bahis.net named.conf

- bahis.net
- 1.168.192.in-addr.arpa
-

root@server:/var/named/master/fz

```
  GNU nano 5.6.1                                    user.net
$TTL 1D
@        IN SOA @ server.user.net. (
               2024072700  ; serial
               1D          ; refresh
               1H          ; retry
               1W          ; expire
               3H )        ; minimum
         NS        @
         A        192.168.1.1
$ORIGIN user.net.
server        A        192.168.1.1
ns        A        192.168.1.1
```

15: bahis.net

- master/fz master/rz
-

```
[root@server.user.net fz]# nano user.net
[root@server.user.net fz]# cp /var/named/named.loopback /var/named/master/rz/
[root@server.user.net fz]# cd /var/named/master/rz/
[root@server.user.net rz]# mv named.loopback 192.168.1
[root@server.user.net rz]#
```

16: master/fz master/rz

-
- $TTL 1D
- SOA-
-
- A- PTR-
- $ORIGIN

17: DNS- 192.168.1

- SELinux-
- named_write_master_zones
- systemctl restart named

```
[root@server.user.net rz]# journalctl -x -f
Feb 07 18:49:27 server.user.net kernel: SELinux:  Converting 624 SID table entries...
Feb 07 18:49:27 server.user.net kernel: SELinux:  policy capability network_peer_controls=1
Feb 07 18:49:27 server.user.net kernel: SELinux:  policy capability open_perms=1
Feb 07 18:49:27 server.user.net kernel: SELinux:  policy capability extended_socket_class=1
Feb 07 18:49:27 server.user.net kernel: SELinux:  policy capability always_check_network=0
Feb 07 18:49:27 server.user.net kernel: SELinux:  policy capability cgroup_seclabel=1
Feb 07 18:49:27 server.user.net kernel: SELinux:  policy capability nnp_nosuid_transition=1
Feb 07 18:49:27 server.user.net kernel: SELinux:  policy capability genfs_seclabel_symlinks=1
Feb 07 18:49:27 server.user.net setsebool[8795]: The named_write_master_zones policy boolean was changed to 1 by roo
t
Feb 07 18:49:33 server.user.net dbus-broker-launch[5973]: avc:  op=load_policy lsm=selinux seqno=4 res=1
Feb 07 18:50:39 server.user.net dbus-broker-launch[665]: avc:  op=load_policy lsm=selinux seqno=4 res=1

Feb 07 18:51:52 server.user.net systemd[5943]: selinux: avc:  op=load_policy lsm=selinux seqno=4 res=1
Feb 07 18:51:52 server.user.net systemd[5943]: Started VTE child process 8803 launched by gnome-terminal-server proc
ess 6885.
    Subject: A start job for unit UNIT has finished successfully
    Defined-By: systemd
    Support: https://wiki.rockylinux.org/rocky/support

    A start job for unit UNIT has finished successfully.

    The job identifier is 616.
```

18: SELinux- named

- dig ns.bahis.net
- NOERROR
- A- 192.168.1.1
- 127.0.0.1#53

```
[root@server.user.net fz]# dig ns.user.net

; <<>> DiG 9.16.23-RH <<>> ns.user.net
;; global options: +cmd
;; connection timed out; no servers could be reached

[root@server.user.net fz]#
```

19: dig ns.bahis.net